

Education

The Hong Kong University of Science and Technology

Ph.D. in Electronic and Computer Engineering; advisor: Prof. Jun Zhang

Hong Kong SAR

Feb. 2025 – Present

Beihang University

B.Eng. in Computer Science and Technology

Beijing, China

Sep. 2020 – Jun. 2024

Research Interests

My research focuses on building efficient and high-performing large-scale AI models. I currently study reinforcement learning for efficient image and video generation; my prior work spans low-precision inference, computation skipping, efficient attention, and model compression for visual and language generative models.

Selected Publications

* equal contribution; † corresponding author.

[Full list](#)

MeanFlowNFT: Bringing Forward-Process RL to Average-Velocity Generators [📄](#) [🗨](#)

Yushi Huang*, Xiangxin Zhou*†, Jun Zhang, Liefeng Bo, Tianyu Pang†

Preprint, 2026.

MoDES: Accelerating Mixture-of-Experts Multimodal Large Language Models via Dynamic Expert Skipping [📄](#) [🗨](#)

Yushi Huang, Zining Wang, Zhihang Yuan†, Yifu Ding, Ruihao Gong, Jinyang Guo, Xianglong Liu, Jun Zhang†

CVPR 2026.

LinVideo: A Post-Training Framework towards $\mathcal{O}(n)$ Attention in Efficient Video Generation [📄](#)

Yushi Huang, Xingtong Ge, Ruihao Gong†, Chengtao Lv, Jun Zhang†

CVPR 2026. Highlight.

QVGen: Pushing the Limit of Quantized Video Generative Models [📄](#) [🗨](#)

Yushi Huang, Ruihao Gong†, Jing Liu, Yifu Ding, Chengtao Lv, Haotong Qin, Jun Zhang†

ICLR 2026. Top 1.3%.

Reinforcing Few-Step Generators via Reward-Tilted Distribution Matching [📄](#) [🗨](#)

Yushi Huang*, Xiangxin Zhou*, Ruoyu Wang*, Chi Zhang, Jun Zhang, Tianyu Pang†

Preprint, 2026.

HarmoniCa: Harmonizing Training and Inference for Better Feature Caching in Diffusion Transformer Acceleration [📄](#) [🗨](#)

Yushi Huang*, Zining Wang*, Ruihao Gong†, Jing Liu, Xinjie Zhang, Jinyang Guo, Xianglong Liu, Jun Zhang†

ICML 2025.

TFMQ-DM: Temporal Feature Maintenance Quantization for Diffusion Models [📄](#) [🗨](#)

Yushi Huang*, Ruihao Gong*, Jing Liu, Tianlong Chen, Xianglong Liu†

CVPR 2024. Highlight.

Temporal Feature Matters: A Framework for Diffusion Model Quantization [📄](#) [🗨](#)

Yushi Huang, Ruihao Gong, Xianglong Liu†, Jing Liu, Yuhang Li, Jiwen Lu, Dacheng Tao

TPAMI 2025.

LLMC: Benchmarking Large Language Model Quantization with a Versatile Compression Toolkit [📄](#) [🗨](#)

Ruihao Gong*, Yang Yong*, Shiqiao Gu*, Yushi Huang*, Chengtao Lv, Yunchen Zhang, Dacheng Tao, Xianglong Liu†

EMNLP Industry 2024.

Flash-VAED: Plug-and-Play VAE Decoders for Efficient Video Generation [📄](#) [🗨](#)

Lunjie Zhu, Yushi Huang, Xingtong Ge, Yufei Xue, Zhening Liu, Yumeng Zhang, Zehong Lin, Jun Zhang†

ICML 2026.

Light Forcing: Accelerating Autoregressive Video Diffusion via Sparse Attention [📄](#) [🗨](#)

Chengtao Lv, Yumeng Shi, Yushi Huang, Ruihao Gong†, Shen Ren, Wenya Wang†

ICML 2026.

SALT: Self-Consistent Distribution Matching with Cache-Aware Training for Few-Step Video Generation [📄](#) [🗨](#)

Xingtong Ge, Yi Zhang, Yushi Huang, Dailan He, Xiaohong Wang, Bingqi Ma, Guanglu Song, Yu Liu, Jun Zhang†

ECCV 2026.

Experience

- Tencent · Hunyuan

Research Intern; mentor: Tianyu Pang

RL for image and video generative models.

Shenzhen, China
Feb. 2026 – Present
- ByteDance · Seed

Research Intern; mentors: Yang Zhang and Zhihang Yuan

Inference acceleration for multimodal LLMs.

Shenzhen, China
Sep. 2025 – Nov. 2025
- Microsoft Research Asia

Research Intern; mentor: Fangyun Wei

Video generation and world models.

Beijing, China
Dec. 2024 – Feb. 2025
- SenseTime Research

Research Intern (multiple terms); mentor: Ruihao Gong

Compression and acceleration for image and video diffusion models and LLMs.

Beijing & Shenzhen, China
May 2023 – Feb. 2026

Projects

- LightCompress: Accurate and Efficient AIGC Model Compression

Core contributor:

- Implemented a broad suite of LLM quantization methods, including QuaRot, GPTQ, SmoothQuant, and OmniQuant.
 - Built an end-to-end toolkit supporting multiple model architectures, evaluation protocols, and inference backends.
 - Documented practical quantization recipes across model families and deployment settings.

700+ GitHub stars
- LightX2V: Light Video Generation Inference Framework

Contributor:

- Contributed to the framework design for adding Wan and HunyuanVideo model support.
 - Integrated quantization, feature caching, and sparse attention features into the framework.
 - Contributed to model distillation for LightX2V.

2.5k+ GitHub stars

Academic Service

- Conference reviewer: NeurIPS, ICLR, ICML, COLM, AAAI, CVPR, and ECCV.

Skills

- Programming & Frameworks: Python, C, Java, PyTorch, and NumPy.
- Research: Reinforcement learning, model compression, and model acceleration.